COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	7/17/2026	**DATE SUBMITTED:**	
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ACTIVITY NO. _2.1_ Arrays, Pointers and Dynamic Memory Allocation

1. PROCEDURE OUTPUT

Table 2-1. Initial Driver Program

Screenshot

```
OA 2.1 > ILO A & B.cpp > main()
#include <iostream>
#include <string.h>

class Student {
  private:
    std::string studentName;
    int studentAge;

//Constructor
  public:
  Student (std::string newName = "Jerry Matthew", int newAge = 19){
      studentName = std::move(newName);
      studentAge = newAge;
      std::cout << "Constructor Called." << std::endl;
  }
//Deconstructor
  ~Student(){
    std::cout << "Destructor Called." << std::endl;
  }
//Copy Constructor
  Student(const Student &copyStudent) {
    std::cout << "Copy Constructor Called" << std::endl;
    studentName = copyStudent.studentName;
    studentAge = copyStudent.studentAge;
  }
//Display Attributes
  void printDetails(){
    std::cout << this->studentName << " " << this->studentAge << std::endl;
  }
};


int main(){
  Student student1("Gerard", 28);
  Student student2(student1);
  Student student3;
  student3 = student2;

  return 0;
}
```

```
Constructor Called.
Copy Constructor Called
Constructor Called.
Destructor Called.
Destructor Called.
Destructor Called.
PS C:\Users\TIPQC\Documents\Larracas_CPE21S4>
```

Observation	The first Constructor called moved a bane “Gerard” from student3 to student1 using move(newName) and newAge to initialize Gerard’s age which is 28. Next output Copy Constructor was called by copying the Student name and its new age and the second

	constructor was called by pasting the copied constructor with its student name and newAge which is “Jerry Matthew” and 19(newAge) to the student3 section. Then the student const points the address of Student to student3 and the pointer of const struct student points to the address of student2 so, they now have the same studentName and studentAge. And lastly the system seems to just repeats the destructor called output without any change except that it removes the name “Gerard” in student1.

Table 2-2. Modified Driver Program with Student Lists

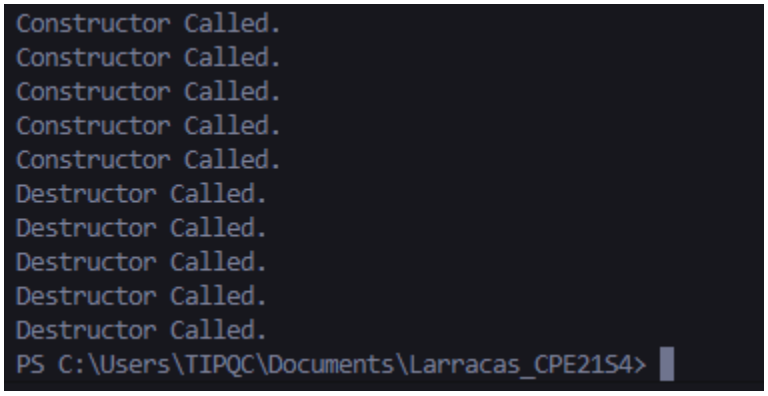
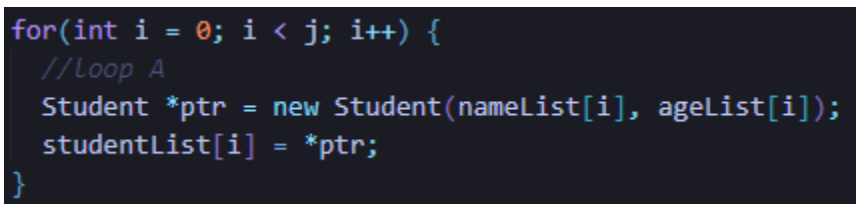
Screenshot	 
Observation	This time we added arrays and the first constructor called is storing the Student name and age in the public constructor at array 0 of namelist and pointing it to studentList array 0. The second, third, fourth and fifth constructor called also did the same as the first constructor. Then the system will now store the names that is being declared in the nameList array and those names are “Carly”, “Freddy”, “Sam”, “Zack” and “Cody”. It will also do the same on the ageList array and will store their ages.

Table 2-3. Final Driver Program

Loop A	
Observation	Loop A creates a for loop and created a pointer with a new student of namelist and agelist but this time it will show the output of the names and ages.

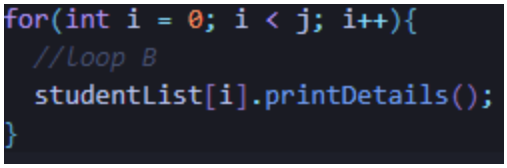
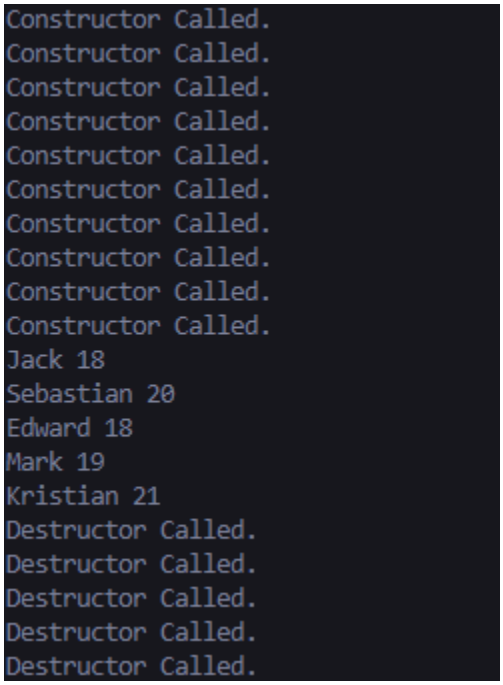
Loop B	
Observation	This part will show the details of the student on the list including their name and age.
Output	
Observation	As we can see the output shows the names that we input and their ages beside them. The Constructor called is also the same as the Table 2-1 observation and destructor called to destroy a certain object and release allocated memory.

Table 2-4. Modifications/Corrections Necessary

Modifications	
Observation	

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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